Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 17)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 1 . 2 Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 1 . 3 Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages .....

.....

.....

.....

Disadvantages .....

.....

.....

.....

Do not write outside the box

10

0 2 Table 1 shows the pH of rainwater and the number of fish species in lakes in four areas of Scandinavia with different bedrock geology.

Table 1

Lake area	Bedrock	Mean rainfall pH	Mean lake pH	Number of fish species	Dissolved aluminium / mg dm^{-3}
A	Granite	4.3	4.6	0	0.42
B	Granite	4.8	5.1	3	0.20
C	Limestone	4.3	7.2	8	0.01
D	Limestone	5.5	7.6	9	0.01

0 2 . 1 State the pH below which rain is described as acid rain and explain why unpolluted rain is already slightly acidic.

[2 marks]

.....

.....

.....

.....

0 2 . 2 Use Table 1 to explain why lakes in area C contain many more fish species than lakes in area A, even though the rainfall has the same pH.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 2 . 3 Describe **two** effects of acid rain on non-living objects.

[2 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

0 2 . 4 Explain how sulfur dioxide released from a power station is converted to sulfuric acid in the atmosphere.

[2 marks]

.....

.....

.....

.....

10

0 3 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 3 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

.....

.....

.....

.....

0 3 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

.....

.....

.....

Do not write outside the box

5

0 4 . 1 Describe how hydrothermal deposition produces exploitable concentrations of metal ores.

[4 marks]

0 4 . 2 Explain how Proterozoic marine sediments produced the world's major iron ore deposits.

[2 marks]

0 4 . 3 Explain how alluvial (physical sediment) processes separate and concentrate minerals such as gold and tin ore.

[2 marks]

0 4 . 4 Give **two** examples of mineral deposits formed by biological sedimentation and state their origin.

[2 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

10

0 5 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 5 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

.....

.....

.....

.....

0 5 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

.....

.....

.....

.....

5

Do not write outside the box

0 6 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 6 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

.....

.....

.....

.....

0 6 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

.....

.....

.....

.....

5

0 7 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

.....

.....

.....

.....

0 7 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

.....

Do not write outside the box

Do not write outside the box

0 7 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

0 7 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

1

2

3

0 7 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

Do not write outside the box

.....

.....

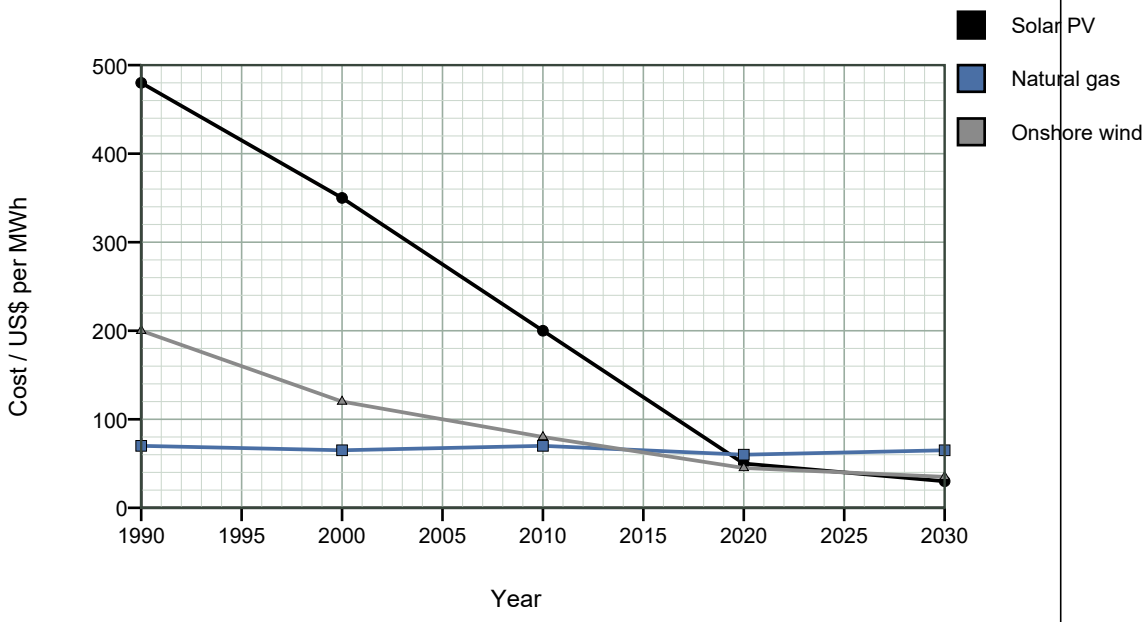
.....

.....

15

0 8 Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



0 8 . 1 Describe the trends shown in Figure 1.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 8 . 2 Explain why new energy technologies are expensive when first introduced and why their costs then fall.

[3 marks]

Do not write outside the box

0 8 . 3 Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

0 8 . 4 Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]

10

0 9 . 1 Explain the locational requirements for harnessing wave power.

[2 marks]

0 9 . 2 Describe how **one** named wave power device generates electricity.

[2 marks]

Do not write outside the box

.....

.....

.....

0 9 . 3 Give **two** design problems that have delayed the development of wave power.

[2 marks]

1 .....

.....

2 .....

.....

0 9 . 4 Compare the environmental impacts of tidal barrages, tidal lagoons and in-stream tidal turbines.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

Do not write outside the box

1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

Do not write outside the box

Extra space

.....

.....

.....

15

Do not write outside the box

Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

1 1 . 2 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS